LoomPilot: Project Document

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1. Executive summary

LoomPilot is an AI-assisted order, quotation and production platform for a garment and textile manufacturing unit. A client submits an order with a deadline. An AI agent checks whether the unit can meet it, using machines, materials, workers and the load already committed. If it can, a second agent prepares a quotation for the admin to approve and send. If it cannot, the admin receives the reasons and recommendations, including local market purchase options, and decides what to do.

A third agent watches raw material prices every day. When prices surge, it proposes revised prices for affected quotations. The admin approves, and the client is notified. A client chat assistant explains prices and revisions, and passes any discount request to the admin. After the client accepts, the admin hands the order to supervisors, who submit daily reports that drive progress tracking for both the admin and the client.

The system is built on the MERN stack with TypeScript, with the AI agents written in Python in an agents folder, and a layered SOLID backend.

2. Background and problem statement

- Quotations are prepared by hand from rough estimates, so orders can be accepted at little or no profit.
- Deadlines are promised without checking real capacity, committed orders, machine availability or attendance.
- Raw material prices move between quotation and purchase, and the quoted price stops being profitable.
- Clients ask why a price changed, and answers depend on the admin's time.
- Progress is not visible. Admins chase supervisors, and clients chase admins.
- Daily resource information (attendance, machines, stock, output) is scattered and not used for planning.

3. Vision and objectives

Vision: every order gets a realistic deadline, a profitable price and visible progress, with the admin making the decisions and the agents doing the analysis.

ID	Objective
O-1	Give an honest feasibility answer for every order, based on real capacity and committed load.
O-2	Produce accurate quotations that protect margin, and prepare them quickly for admin approval.
O-3	Explain why a deadline cannot be met and recommend ways to meet it, including local purchases.
O-4	Detect raw material price surges daily and re-price affected orders in a controlled, approved way.
O-5	Let clients ask questions and receive consistent answers, while discounts stay with the admin.
O-6	Make daily resource and production data available through a simple supervisor portal.
O-7	Give admin and clients live visibility of each order's phase and produced quantity.
O-8	Raise on-time delivery and reduce admin time spent on estimation and status questions.

4. Users and stakeholders

Role	Description	Key needs
Admin (owner or manager)	Runs the business side, may be several users	Fast, safe approvals, clear reasons and options, control of price and discounts, full visibility
Supervisor	Runs production and reports daily	Simple mobile reporting, clear assigned tasks

Role	Description	Key needs
Client	Buys from the manufacturer	Easy ordering, clear quotes, honest deadlines, answers, progress visibility
Suppliers	Indirect	Purchase orders and delivery schedules
Project and development team	Builders	Clear scope and contracts

5. Scope

5.1 In scope for the first release

- Three portals: client, admin and supervisor, with role-based access.
- Client order submission, quotation view, accept or decline, chat and progress tracking.
- Feasibility analysis by agent, using machines, materials, workers, attendance and committed load.
- Quotation drafting by agent, admin review, approval, PDF and sending.
- Recommendations for infeasible orders, including local-market purchase options and prices.
- Admin decisions: buy materials, propose an alternative, or decline.
- Daily market price monitoring, surge alerts and price revisions with admin approval.
- Client chat assistant with escalation of discount requests to the admin and admin takeover.
- Order handover from admin to supervisors.
- Supervisor daily report: attendance, machines, material stock, production.
- Production tracking and a client-safe progress view.
- Basic delivery, invoice and receipt recording.
- Dashboards, reports, notifications, audit and settings.

5.2 Out of scope for the first release

- Full ERP functions such as accounting, payroll and tax filing.

- Automatic purchasing or automatic hiring. The system recommends and the admin decides.
- Online payment collection and automated payment reminders.
- Machine sensor (IoT) integration.
- Granting discounts automatically, or any AI action that changes a price without admin approval.
- Multiple manufacturing units.

6. Functional requirements

Priority: H is high, M is medium, L is low.

6.1 Client portal

ID	Requirement	Priority
CP-1	The client can log in to an account created by invitation from the admin.	H
CP-2	The client can submit an order with product, quantity, deadline, specifications, delivery address and reference files.	H
CP-3	The client can see the status of every order they placed.	H
CP-4	The client can view an approved quotation (price, delivery date, validity, terms) and download its PDF.	H
CP-5	The client can accept or decline a quotation or a revised price.	H
CP-6	The client can chat about an order, and receive replies from the assistant or the admin.	H
CP-7	The client can see a progress timeline with the current phase, produced and remaining quantity and estimated completion date.	H
CP-8	The client receives notifications for quotation sent, revised price, progress milestones and delivery.	M

ID	Requirement	Priority
CP-9	The client sees client-safe information only, with no cost or margin.	H

6.2 Admin portal

ID	Requirement	Priority
AD-1	The admin sees an inbox of new orders with their feasibility status.	H
AD-2	The admin sees each feasibility report: verdict, ETA, bottleneck, risk factors, reasons and options.	H
AD-3	The admin can review, edit, approve or reject a quotation, with a margin warning.	H
AD-4	The admin can send an approved quotation to the client.	H
AD-5	For at-risk or infeasible orders the admin can buy materials, propose an alternative or decline.	H
AD-6	The admin is notified when a client accepts, and can hand the order to one or more supervisors.	H
AD-7	The admin can track every order by phase and see progress, daily reports and alerts.	H
AD-8	The admin sees price alerts, trend charts and the orders affected by a surge.	H
AD-9	The admin can review, edit, approve or reject proposed price revisions.	H
AD-10	The admin sees all chats, receives discount escalations, can reply and can take over a conversation.	H
AD-11	The admin manages clients, users, materials, machines, workers, products, suppliers and settings.	H

ID	Requirement	Priority
AD-12	The admin can view the audit log and the agent activity log.	M
AD-13	The admin can record deliveries, invoices and receipts.	M

6.3 Supervisor portal

ID	Requirement	Priority
SV-1	The supervisor sees the orders assigned to them with plan and due dates.	H
SV-2	The supervisor submits one daily report covering attendance, machine availability, material stock and production per order and process.	H
SV-3	The form works on a phone and offline, and syncs when a connection returns.	H
SV-4	The supervisor can correct a report with a reason.	M
SV-5	The supervisor can raise an issue (breakdown, shortage, delay).	M
SV-6	The supervisor is reminded when the daily report is missing.	M

6.4 Feasibility, quotation and procurement

ID	Requirement	Priority
FQ-1	The system calculates labour and machine capacity per process and uses the lower value.	H
FQ-2	The system subtracts the remaining load of confirmed orders from capacity.	H
FQ-3	The system checks material availability including wastage, and uses supplier lead time for shortfalls.	H

ID	Requirement	Priority
FQ-4	The system produces a verdict (feasible, at risk, not feasible), an ETA and risk factors with severity.	H
FQ-5	For infeasible orders the system proposes options with their effect on ETA and price: overtime, extra workers, later date, partial shipment, local purchase.	H
FQ-6	Local purchase options list suppliers or market sources with price and lead time.	H
FQ-7	The quotation includes itemised cost, wastage, price-risk buffer, margin, validity, ETA basis, and an explanation.	H
FQ-8	A purchase decision by the admin is recorded, creates a supplier order and triggers a new feasibility run.	H
FQ-9	Quotation terms include a price-variation clause.	H
FQ-10	Quotations expire after their validity period.	M

6.5 Market price monitoring and revisions

ID	Requirement	Priority
MP-1	The admin manages a watchlist of materials and market sources.	H
MP-2	The agent collects prices daily and stores each observation with source, time and verification status.	H
MP-3	The agent computes daily change, seven-day change and volatility, and compares them with configurable thresholds.	H
MP-4	A surge creates an alert and identifies affected orders, excluding materials already purchased.	H

ID	Requirement	Priority
MP-5	The agent proposes a price revision with old price, new price, percentage change, drivers, date impact and a client-friendly reason.	H
MP-6	A revision reaches the client only after admin approval.	H
MP-7	The client can accept, decline or ask about a revised price.	H
MP-8	Large moves from unverified sources require admin confirmation.	M

6.6 Chat assistant

ID	Requirement	Priority
CH-1	Each order has a conversation between the client, the assistant and the admin.	H
CH-2	The assistant explains prices, price changes, deadlines and progress using client-safe approved data.	H
CH-3	The assistant never grants discounts, changes price or dates, or reveals cost and margin.	H
CH-4	A discount or extra request creates an escalation to the admin with a summary.	H
CH-5	The admin can reply, issue a revised quotation through the revision flow, or take over the conversation.	H
CH-6	Messages are labelled by sender, and figures in assistant replies are checked against records.	H

6.7 Production tracking and platform

ID	Requirement	Priority
PT-1	The system derives phase, percent complete, produced and remaining quantity and projected finish from daily reports.	H
PT-2	The system raises a behind-schedule alert with recovery suggestions.	M
PL-1	Role-based access for admin, supervisor and client, with client data isolation.	H
PL-2	Every approval, decision and agent output is audited.	H
PL-3	Alerts and notifications go out in-app and by email, with retry.	H
PL-4	Settings cover thresholds, margins, wastage, risk buffer, validity, safety days, calendar and watchlist.	H
PL-5	Reports: order pipeline, on-time delivery, price-change impact, quoted versus actual.	M
PL-6	CSV import of materials, workers, machines, products and price history.	M

7. Non-functional requirements

Area	Requirement
Usability	Simple client portal, mobile-first supervisor forms with offline entry
Performance	Feasibility and quotation typically ready within 60 seconds, with live status
Reliability	Retries for notifications and agent calls, no lost events, failed runs can be retried

Area	Requirement
Security	Encryption in transit and at rest, role-based access, client data isolation, hashed tokens, login lockout and rate limits
Privacy	Clients see only their own client-safe data, and the chat agent can read only client-safe data
Explainability	Every feasibility report and quotation shows its inputs, assumptions and agent run details
Scalability	Stateless API and agents, queue-based processing, tenant-ready data model
Availability	Backups, health checks, graceful shutdown
Language	English first, structure ready for local languages
Auditability	Immutable audit trail of changes and agent actions

8. Technology stack

Layer	Technology
Frontend	React, TypeScript, Vite, Tailwind CSS, shadcn/ui, TanStack Query, React Hook Form, Zod, Recharts, PWA
Backend	Node.js, Express, TypeScript (strict), layered SOLID architecture (routes, controllers, services, repositories, models, middleware, utils), dependency-cruiser
Database	MongoDB replica set with Mongoose
Queues and realtime	BullMQ on Redis, Socket.IO
AI agents	Python 3.12, FastAPI, Pydantic v2, in the agents folder; Anthropic API behind a provider interface; NumPy, pandas, SciPy for calculations

Layer	Technology
Integrations	SMTP email first, WhatsApp and SMS through provider adapters, market price source adapters, pdfmake for PDFs
Quality	Vitest, Supertest, mongodb-memory-server, pytest, ruff, mypy, Playwright, OpenAPI and Swagger
Delivery	pnpm monorepo, Docker, GitHub Actions, built in stages with Antigravity and reviewed at checkpoints

The full design is in the System Architecture Document.

9. Data requirements

Inputs: client orders and specifications, products with bill of materials and routing, materials and stock, machines and availability, workers with daily attendance and output, supplier details, lead times and quotations, daily market prices, holidays and settings.

Outputs: feasibility reports, quotations and revisions, price alerts, chat messages and escalations, production plans, progress snapshots, deliveries, reports.

Data needed to start: a list of machines by process with capacity per hour, standard production times per product and stage, current workers with typical attendance, current stock and material rates, supplier contacts with lead times, and ten to thirty past orders with actual times and costs for testing the engine.

10. Decisions made

#	Decision
D-1	Three roles: admin, supervisor, client.
D-2	The admin approves every quotation and every price revision before the client sees it.
D-3	After a price surge, the admin approves the revised price first, then the client is notified.

#	Decision
D-4	The chat assistant explains and handles client questions and negotiation by explanation. Any discount or extra demand is sent to the admin, who decides.
D-5	When a client accepts, the admin is notified and hands the order to supervisors. There is no advance payment gate.
D-6	Stack: MERN with TypeScript, Python agents in the agents folder, layered SOLID backend.
D-7	Agents only propose. Only the Node API writes data.
D-8	Five agents: feasibility, quotation, market price, client assistant, and monitoring (recommended).

11. Assumptions

Please correct any that are wrong.

1. The unit makes garments and textiles. Production stages are configurable per product, with cutting, stitching, finishing and packing as the sample garment routing.
2. One manufacturing unit in the first release.
3. Clients are invited by the admin, with no open sign-up.
4. Machines are tracked individually with a type and capacity per hour.
5. At least one automated market price source will be available, in addition to supplier quotations entered by the admin.
6. The first release records payments but has no online payment and no automated reminders.
7. The admin role can be held by several users. There are no separate sales, store or accounts roles.
8. Supervisors enter data by hand through the portal.
9. Email and in-app notifications first, WhatsApp and SMS afterwards because they need provider approvals.
10. The chat is in English first.

11. The admin cannot send a quotation for an order marked not feasible without choosing an option such as a purchase or an alternative.

12. Constraints

- Prediction quality depends on the quantity and quality of daily data from supervisors.
- Market data depends on available sources and their reliability.
- AI features depend on an external LLM API, with cost and availability limits.
- Budget and timeline are to be confirmed by the sponsor.
- Price-variation terms and client communication wording need legal review.

13. Risks and mitigation

Risk	Impact	Mitigation
Poor or sparse capacity data at the start	Wrong feasibility answers	Conservative buffers, low-data warnings, backtesting on past orders, improving as data grows
Supervisors do not submit daily reports	Capacity and tracking go stale	Simple offline form, reminders, missing-report flag, last-known data marked stale
Unreliable market price data	False surges or missed surges	Supplier quotes as baseline, multiple sources, unverified flag and admin confirmation
Clients dispute price increases	Lost trust, legal issues	Price-variation clause, clear client-friendly reasons, admin approval, locked prices for purchased materials
Chat assistant states something wrong	Commitments or confusion	Client-safe data only, figure checking, no authority to change terms, escalation when unsure
Over-reliance on AI	Wrong decisions	Admin approves all outputs, every output shows its inputs

Risk	Impact	Mitigation
LLM cost or outage	Delays	Calculations run without the LLM, fake provider for tests, budgets, retries, failed runs can be retried
Scope growth	Delays	Fixed release scope, deferred list, change control
Low adoption	Benefits lost	Pilot with real orders, training, involving admin and supervisors early

14. Success criteria

Targets are proposed and should be confirmed against a baseline from the unit's recent orders.

Measure	Proposed target
Orders delivered by the promised date	Improvement over baseline, for example 15 percentage points after three months
Admin time to produce a quotation	Under 15 minutes of admin effort for a standard order
Feasibility verdict accuracy	Most feasible verdicts delivered on time, tracked per month
Supervisor daily report submission	At least 90 percent of working days
Price surge detected and revision proposed	Same day as detection
Client questions answered by the assistant without admin help	At least 70 percent, excluding discount requests
Quoted versus actual cost variance	Reduced compared with baseline

15. Delivery plan

Indicative durations, to be confirmed.

Phase	Scope	Weeks
0. Discovery	Requirements, data collection, garment routing and standard times, market source choice	2
1. Foundation	Monorepo, platform conventions, auth and three roles, security	3
2. Resources	Products and bill of materials, materials and stock, machines, workers, supervisor daily report	4
3. Order to quotation	Orders, feasibility agent, quotation agent, admin and client screens	6
4. Market prices	Market price agent, alerts, revisions and approval flow	4
5. Chat assistant	Conversations, assistant, escalation, takeover	3
6. Tracking and delivery	Monitoring agent, progress views, delivery, payments, reports	3
7. Testing and pilot	End-to-end tests, pilot on real orders, tuning	4
8. Go-live	Deployment, training, support	2

The total is about 31 weeks if done in sequence, and roughly 24 to 28 weeks when frontend and agent work run in parallel.

Suggested team: two full-stack MERN developers, one Python and LLM developer, one frontend and UI developer with part-time design, a part-time tester, and a product owner with one person from the unit who knows the processes.

Review points: after the foundation, after the order-to-quotation phase (the core of the product, where the formulas are checked against real past orders), and before the pilot.

16. Acceptance criteria

- A client can submit an order and see its status.
- A feasibility report is produced with verdict, ETA, reasons and risk factors.
- For a feasible order a quotation is drafted, approved by the admin and received by the client.

- For an infeasible order the admin sees reasons and options, including local purchase prices, and can buy, propose an alternative or decline.
- A simulated price surge produces an alert and a proposed revision, and the client is notified only after admin approval.
- The assistant answers a price question from approved data and escalates a discount request to the admin.
- Accepting a quotation notifies the admin, who can assign supervisors.
- Supervisor daily reports update progress that the admin and the client can see.
- Role isolation holds: a client cannot see another client's data or any cost or margin.
- The pilot is signed off on real orders.

17. Open questions

- Which market price sources and any paid feeds are acceptable?
- Which garment stages, standard times and machine details apply to your unit?
- Should the admin be able to proceed with an at-risk order and accept the risk on record?
- Should clients log in with a password or a one-time code?
- Is an advance payment step needed in a later release?
- Budget, timeline and hosting preferences.

18. Glossary

Term	Meaning
Feasibility	Whether the unit can deliver an order by the client's deadline
Committed load	Remaining work of orders already confirmed
Capacity	Output per day a process can produce, limited by workers and machines
ETA	Estimated completion date calculated from materials, capacity and safety days

Term	Meaning
Price-risk buffer	Extra cost allowance for possible raw material price increases
Surge	A raw material price move beyond the configured thresholds
Revision	An approved change to a quotation's price or date, with a reason
Escalation	A chat request the assistant passes to the admin
Client-safe	Information suitable for a client, with no cost or margin
Agent	An AI component that analyses data and proposes outputs for human approval

19. Approvals

Name	Role	Signature	Date
	Project sponsor		
	Project manager		
	Business owner		